Bootstrap Confidence Intervals

Jericho Lawson

MATH 240

Week 10, Lecture #2 — Ch. 12

Reminders

- Feedback given on proposals!
- Read through Chapter 12 of IMS over the next few days
- **Checkpoint #4** on Friday!
  - Open-note and done individually
  - Focused on hypothesis testing and confidence intervals
    - ★ You will not be writing R code, but you may need to interpret results from R

Ways to Estimate p

- A **point estimate** is a single plausible value for a parameter.
  - e.g. $\hat{p}$
- A **confidence interval** is a range of plausible values for a parameter.

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Question: Why do we need a confidence interval if we already have a point estimate? *Shows the uncertainty (i.e., sampling variability) associated with a point estimate.*

Confidence Intervals

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This idea is to create a bootstrap sampling distribution of your test statistic and then a 95% confidence interval is an interval of the middle 95% values of bootstrap sampling distribution.

Confidence Intervals with Bootstrapping

Recall bootstrapping from last week:

- Use: Conduct same experiment with sample from null distribution
- The difference here: we will treat the sample as the actual population
  - Sampling with replacement will occur here
  - Provides us with different point estimates, for which we get a $100 \times (1 - \alpha)\%$ confidence interval for

Steps for Bootstrap CI

Begin with the sample we actually collected. Let n be the sample size (number of observations).

1. From original sample, sample n observations with replacement. This is your bootstrap sample.
2. Calculate the statistic in your bootstrap sample. This is your bootstrap statistic.
3. Repeat the steps many times ($B \approx 10,000$). The distribution of bootstrap statistics is your bootstrap sampling distribution! It serves as a model for the actual sampling distribution.

Constructing CIs

Once you have your bootstrap sampling distribution, you can get a (bootstrap percentile) confidence interval as follows:

1. Take the $(1 - \alpha/2)$%th percentile of values from your boostrap distribution (e.g. 2.5th for 95% CI)
2. Take the $100 \times (1 - \alpha/2)$%th percentile of values from your bootstrap distribution (e.g. 97.5th for 95% CI)

These two values form the endpoints of your confidence interval.

Sampling Distribution

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- Testing a hypothesis!
  - Sampling distribution was generated assuming null is true (i.e. null sampling distribution)
- Here, we are estimating the unknown value of a parameter, which means we need the **sampling distribution**

Example: Public Transit

Suppose we were curious about whether those in a metropolitan area were taking public transit as much as what is seen nationally. We decide to take a sample of 150 from a population of 500,000.

With this sample, we will sample 150 from this sample repeatedly with replacement, meaning we may get duplicate subjects in our samples. From there, we might find that:

1. 4 out of 150 take public transit
2. 8 out of 150 take public transit
3. 13 out of 150 take public transit
4. 7 out of 150 take public transit
5. ...and so forth

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We take the $(1 - \alpha/2)$th and $100 \times (1 - \alpha/2)$%th percentiles as the bounds of our $100 \times (1 - \alpha)$% confidence interval:

- e.g. 95% CI: (0.0400, 0.1267)
- Interpretation: We are 95% confident that the true proportion of people taking public transit in this metropolitan area is between 0.04 and 0.1267.

Example: Voting Among Young Americans

The 53rd Harvard Youth Poll, conducted September 30 - October 7 among 2,024 young Americans, finds a widening partisan gap in voting intention. Half of young Democrats say they will definitely vote in November, compared with 34% of young Republicans—a 16-point gap, up from five points in 2022 and 11 points in 2018. Overall, 33% of young Americans say they will definitely vote, compared with 40% who said the same ahead of both the 2018 and 2022 midterms.

Figure: Source: Harvard Youth Poll

Example: Bootstrap CI

1. Note the following items:
   - Sample size $n = 2024$, $\hat{p} = 0.33$
2. For $B = 10000$ samples: sample 2024 subjects with replacement, calculate $\hat{p}_i^*$
3. Find 2.5th and 97.5th percentiles of all $\hat{p}_i^*$, resulting in bootstrap confidence interval

Let's look at the code to see how it works!

Example: Interpreting CIs

Recall p is the true proportion of all young Americans who are definitely voting.

In our bootstrap sample, the 95% confidence interval for p is (0.3103, 0.3503).

- We are 95% confident that the true percentage of all young Americans who are definitely voting is between 31.03% and 35.03%
- Recall: in the long run (of computing 95% CIs many, many times), we find that 95% of 95% confidence intervals contain the true population parameter.

Activity: Run your own Bootstrap!

In groups of 2-3, go to the following site: https://www.sciencedaily.com/.

- Find a study in there that discusses a percentage of subjects (or a sample mean) having some result.
- Find a bootstrap confidence interval with the given information, using the R code provided in the last example.
- Report your findings.

Summary

So far we have introduced hypothesis testing and confidence intervals. Here are the approaches we've seen so far (and will get to):

- **Bootstrap** - sampling with replacement
- **Randomization** - shuffling group assignments
- Upcoming: **Mathematical Models**

Summary

1. The parameter α has shown up in two places:
   - In setting the confidence level (i.e., desired coverage probability) of our $100 \times (1 - \alpha\%$ confidence intervals
   - In setting the discernibility level α of our hypothesis tests
2. We also have the following understanding of what confidence intervals and hypothesis tests tell us:
   - The confidence interval defines a range of plausible values for our unknown parameter that are consistent with the observed data
   - Our hypothesis test rejects H_0 if the observed data are sufficiently inconsistent with the null

Summary

Confidence intervals and hypothesis testing are two sides of the same coin!

1. For a 95% CI for p, we consider a two-sided hypothesis test of $H_0 : p = 1/6$ vs. $H_a : p \neq 1/6$:
   - Null parameter value is not in 95% CI $\implies$ reject H_0 at $\alpha = 0.05$ level
   - Null parameter value is in 95% CI $\implies$ fail to reject H_0
2. Suppose we have a two-sided hypothesis test with $\alpha = 0.05$:
   - Test rejects H_0 $\implies$ 95% CI excludes null parameter value
   - Test fails to reject H_0 $\implies$ 95% CI includes null parameter value

So Far...

	Confidence Intervals	**Hypothesis Tests**
Guiding Question	What is a range of plausible values for p that are consistent with the data?	Is the observed data consistent with H_0?
Distribution	Sampling Distribution	Null Distribution
Process	Sample with replacement from the data	Generate simulated data from the null model
Summary Statistic	Point estimate $\hat{p}$	Test statistic $\hat{p}$
Level	$100 \times (1 - \alpha)\%$	α
Result	Interval centered at the point estimate	p-value

Use of Mathematical Models

So far, we've explored hypothesis testing through the use of computational techniques. We can also use models in order to conduct a test, mainly through the use of the Normal distribution.

